1

Introduction

Automation: designing, building and implementing automatic machines. How can such an intriguing concept that has the potential to keep manufacturing located domestically also cause some to be so concerned? The image of intelligent machines producing thousands of quality products for less cost is the dream of many an engineer. But to the current worker at the plant who might be replaced, automation is a potential nightmare.

Depending on one's role in this world, the impact of automation can cause excitement or fear. Let us look at some of these roles to gain an initial view before we start to think about designing and building an automatic machine:

- *Manufacturing Director* — International competition continues to pressure almost all manufacturing operations to reduce production costs. Because labor costs rarely go down, and many workers are operating at reasonably optimal rates, there are no significant gains to be made. Automation, if possible, is a goal of most Manufacturing Directors to remove labor and increase output and quality. Internal manufacturing means greater control of production.
- *Company CEO* — In addition to the concerns of the Manufacturing Director, the CEO is also concerned about employee injuries, Workman's Compensation costs, and the flexibility to raise or lower production outputs if market conditions change. Many a CEO would like to keep production onshore rather than move it to a Third World nation.

Copyright © 2005 by Marcel Dekker

- *Stockholders* — To be viable, an investment needs to be placed in a company that has room to grow when markets increase. Adding automation is often easier than finding skilled employees willing to work third shift. So a company that is highly automated can be perceived to be a good long-term investment.
- *Current Company Worker* — They have the most reason to be concerned with the implementing of automation. Some may lose their jobs, while others may be retrained and relocated to maintain the automated production. Fully automated, or "lights out" facilities, are not always cost-effective. However, the jobs that will be replaced are often ones that seem like drudgery and lead to repetitive motion injuries or other physical risks. In society as a whole, there are hopefully better jobs to be done that require the intelligence of a human being, but remaining competitive means staying in business.
- *Sales Representatives* — If a machine is to be sold to many customers, a set of independent representatives are often formed; or it could be your own company's sales force. Your new automation must generate enough cost savings to make the sale. It needs to be a better mousetrap at a good price to be selected over the competition.
- *Consumer* — The consumer wants high-quality goods at a low price. The social concerns of where and how it is made are often left behind when one gets to the checkout line. However, not all products made halfway around the world are of the quality one would expect. Automation, if done properly, can help to lower costs while keeping or improving quality.
- *Environmentalist* — The product and its packaging will have an impact on the world, perhaps for many years after its useful life is over. How it can be recycled is a concern. Additionally, the automatic machine itself needs to be thought about. A machine that will make only a single size and type of product might be obsolete before it has been assembled and debugged, with today's quickly changing markets. Can a machine be designed such that while still being cost-effective, it can live on to make future products not yet conceived? Can it be designed and made in modules that can be reassembled as building blocks? Or will the one-product machine be limited to being a boat anchor in the not so distant future?

Although the engineer responsible for automation may not hold many of these roles, the concerns of these groups do help to define the dynamics that will come into play either directly into the engineer's world, or indirectly behind the scenes. And if the engineer understands all of these issues, one can better anticipate them and be prepared.

Copyright © 2005 by Marcel Dekker

In this textbook, we will be primarily focused on the traditional engineering aspects of designing and building an automatic machine. The information in the book comes from many sources: companies that manufacture machines, trade show displays and product literature, professional articles and trade magazines, conferences, and the experiences of the author over the past 25 years. These experiences include being a university professor, having a sabbatical at a local automation company, and being the president of two startup companies in automation.

1.1. WHY AUTOMATE?

As the above role descriptions start to mention, there are many reasons to automate a manufacturing process. These include:

- Reducing labor;
- Avoiding labor's sick days, lunch breaks, being late for work;
- Improving quality;
- Reducing waste;
- Enabling production of multiple shifts and weekends;
- Increasing repeatability and quality;
- Increasing Workman's Compensation claims and expenses;
- Keeping production onshore.

Some of these reasons have costs impact savings that are simple to compute. Counting up the X number of employees making $Y per hour can be entered into a spreadsheet. But it is more difficult for a company to put a price tag on improved quality, or to estimate the reduced Workman's Compensation claims. Different accounting practices and methods also create different answers.

Since every manufacturer probably desires to address some, if not all, of these reasons, one could assume that automation is an obvious answer. However, there are not available today automation solutions for every production process. Many production tasks have not yet had a machine designed and made for them. Or the existing machine is not cost-effective compared to the current labor expenses, both direct and indirect.

So the task of automation is often thrust upon a single engineer or engineering team. In some cases there are ready solutions, where the engineer's job is simply to find the best one at the right price. This is similar to what a consumer usually does to buy his or her home washing machine. The consumer rarely designs and builds their own. But in automation, the engineers may need to design and build one from scratch.

If the engineer (or hopefully team) has the skill set and experience, they can do just that, design and build a piece of custom automation. In this case this text is designed to assist the team, either to bring along a novice, or to give more

Copyright © 2005 by Marcel Dekker

experiences to assist the expert. It may be helpful as a reference to justify the methodology.

Alternatively, the team may be charged to outsource the automation; to be responsible to define the system requirements, write a request for proposal, accept bids, select the right bid, and monitor the development and installation process. In this case the text will assist the engineer to understand the basics, appreciate the range of solutions and the level of difficulty, and aid in the evaluation process so as to select a winning bid. Or it may help the engineer to come to the logical conclusion that the process cannot be economically automated and must remain manual labor for the time being.

As we will see, the field of designing automation is both logical and developed, and as creative and individualistic as fine art. There are often many solutions, no one "right" answer, and sometimes no answer at all. But there is often the better mousetrap waiting to be invented, sometimes by the untrained novice who does not know any better why it had not been done before.

1.2. BOOK TOPICS

We will look at the entire process of designing and building automation (Chapters 2–10), including some experiences of the author, both good and not so good. The topics include the traditional areas, including:

- Steps to Automation;
- Justifying Automation;
- The Automation Design Process;
- Robotics as Automation Tools;
- Workstations;
- Feeders and Conveyors;
- Actuators;
- Sensors;
- Control.

Of particular note is the Steps To Automation chapter, where the understanding of the process to be automated is discussed in great detail. Many a machine was designed and built without a good fundamental investigation of how it was to perform the process, and thus never had a chance of succeeding. Examples of both good designs and "war stories" of limited or complete failures will be addressed. Looking at alternative methods to perform the task may make or break the entire machine development program. For example, no one to date has developed a cost-effective alternative to the dexterity of the human hand.

These chapters of the book have been developed as a mix of examples, case studies, thought-provoking questions, and possible individual or student team projects. It has been found that university students learn these concepts best by

Copyright © 2005 by Marcel Dekker

working with them within the framework of a specific project. Most likely time and cost limitations limit a student to only building simple prototypes of key automation process areas, to further gain more understanding on if and how things might work. The degree to which a project is completed by the reader can be a function of time, education to date, and the ultimate goals of the engineer.

Many example projects are the results of previous engineering student teams. Some of these results are promising, while others might be limited. But the author makes note of any issues and limitations, since much can be learned from looking at not so great solutions as well as the great solutions.

Another area of note is the chapter on Bringing New Automation To Market. This is based on the experiences of the author about his three new inventions, from concept, development, system debugging, and market conditions. A rapidly changing economy makes these efforts troubling at times. Patenting automation is also discussed for its benefits and limitations in the chapter on Justifying Automation.

The chapter on System Specifications will be useful for the engineers who must write a Request For Quote (RFQ), understand the submitted proposals, compare alternate design, and look at project management issues. This is useful to people from both sides of the fence, so as to better understand the partnership.

The last chapter, on Packaging Machines, is an introduction to a large subset of automation machines. Packaging is the placing of the finished product into a bag or box, perhaps also into a larger carton, and maybe finally into a cardboard shipping container or box. Whereas many of the processes to be automated are very specific to a very few industries, the packaging market transcends many industries, and thus has a larger market, leading to a greater number of existing machines available off the shelf.

1.3. CONCEPTUAL DESIGN AND THIS TEXT

This text will focus on the concepts relating to the design of automation. The methodology of how a machine is designed is more important than how many bolts are holding it together. The book will also lead the reader towards many existing bodies of knowledge found in more basic engineering textbooks. It will hopefully remind the reader or point to these principles (such as calculating the rotational moment of inertia for motor sizing) but will not develop these concepts within. It is anticipated that the reader has access to these other textbooks as resources.

The figures in this text have been created as a means to show a generalized concept, not a blueprint to be copied into your CAD design. These figures are meant to show how something happens, since it is often difficult to gain such understanding from detailed photographs. Some figures are generalizations of

Copyright © 2005 by Marcel Dekker

many different brands of similar automation machines or devices, rather than a reflection of a specific brand of conveyor or robot. In these changing economic times, it seems more prudent to not focus on any particular brand of automation. So there are not a large number of photographs included. The reader is referred to the Web to search on suggested topics for state-of-the-art components. The author has used this method quite successfully for more than a decade of teaching automation students.

Copyright © 2005 by Marcel Dekker